

# EXPERIMENT 22

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## Problem Statement

Write a menu-driven C program that accepts two positive integers **m** and **n**, where  $m \leq n$ . The menu must: (1) print multiplication tables of **m** and **n** side by side for multipliers 1 to 10, (2) display a multiplication grid for integers 1 to **m** and 1 to **n**, (3) for each integer from **m** to **n** print that integer followed by the next five consecutive integers, and (4) exit. The menu repeats until Exit is selected and invalid choices produce an error message.

## Algorithm

1. Start.
2. Read positive integers **m** and **n**. Validate that  $m > 0$ ,  $n > 0$  and  $m \leq n$ ; otherwise request them again.
3. Repeat the following menu until the user chooses 4:
  - 1 - Multiplication tables of **m** and **n** side by side
  - 2 - Multiplication grid from 1..**m** and 1..**n**
  - 3 - Consecutive-number rows from **m** through **n**
  - 4 - Exit
4. Read choice and use switch(choice).
5. For choice 1, loop **i** from 1 to 10 and print  $m*i$  and  $n*i$  on the same row.
6. For choice 2, use nested loops: rows  $i=1..m$  and columns  $j=1..n$ ; print  $i*j$ .
7. For choice 3, for each  $i=m..n$ , print **i**, **i+1**, **i+2**, **i+3**, **i+4**, **i+5**.
8. For choice 4, terminate the loop.
9. For any other choice, print an invalid-choice error.
10. Stop.

## Test-Case Table

| TC | Input / condition      | Menu choice | Expected result                                             |
|----|------------------------|-------------|-------------------------------------------------------------|
| 1  | m=2, n=5               | 1           | Tables of 2 and 5 printed side by side for 1..10.           |
| 2  | m=2, n=4               | 2           | 2 x 4 multiplication grid containing $i*j$ values.          |
| 3  | m=3, n=5               | 3           | Rows: 3 4 5 6 7 8; 4 5 6 7 8 9; 5 6 7 8 9 10.               |
| 4  | m=1, n=1               | 1           | Boundary valid case; two identical tables of 1 are printed. |
| 5  | m=0, n=4 then m=2,n=4  | -           | Reject m=0; accept the next valid pair.                     |
| 6  | m=5, n=3 then m=3,n=5  | -           | Reject because $m > n$ ; accept the next valid pair.        |
| 7  | m=-2, n=3 then m=1,n=3 | -           | Reject negative input; accept positive values.              |
| 8  | m=2, n=3               | 9           | Print "Invalid menu choice" and show menu again.            |
| 9  | m=2, n=3               | 4           | Exit program normally.                                      |

## C Program

```
#include <stdio.h>

int main(void) {
    int m, n, choice, i, j;

    do {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n);
        if (m <= 0 || n <= 0 || m > n)
            printf("Invalid input. Enter positive values with m <= n.\n");
    } while (m <= 0 || n <= 0 || m > n);

    do {
        printf("\nMENU\n");
        printf("1. Multiplication tables of m and n side by side\n");
        printf("2. Multiplication grid\n");
        printf("3. Next five consecutive integers\n");
```

```

printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        printf("\nTable of %d\t\tTable of %d\n", m, n);
        for (i = 1; i <= 10; i++) {
            printf("%d x %d = %d\t\t%d x %d = %d\n",
                m, i, m*i, n, i, n*i);
        }
        break;

    case 2:
        printf("\nMultiplication Grid:\n\t");
        for (j = 1; j <= n; j++) printf("%4d", j);
        printf("\n");
        for (i = 1; i <= m; i++) {
            printf("%d\t", i);
            for (j = 1; j <= n; j++)
                printf("%4d", i*j);
            printf("\n");
        }
        break;

    case 3:
        printf("\nConsecutive integers:\n");
        for (i = m; i <= n; i++) {
            for (j = 0; j <= 5; j++)
                printf("%d ", i+j);
            printf("\n");
        }
        break;

    case 4:
        printf("Exiting program.\n");
        break;

    default:
        printf("Invalid menu choice. Please enter 1 to 4.\n");
}
} while (choice != 4);

return 0;
}

```

## Compilation and Execution

```

gcc experiment22.c -o experiment22
./experiment22

```

## Sample Verification

**Example:** m = 2, n = 3, choice = 3.

Output rows are:

```

2 3 4 5 6 7
3 4 5 6 7 8

```

This confirms that each integer from m through n is followed by its next five consecutive integers.

## Result

The menu-driven C program was prepared using loops and a switch statement. It validates m and n, performs all three required operations, handles invalid menu choices, and continues until the Exit option is selected.